

NOTE: Apparently the uni printers give up as soon as I use a different coloured pen, or a light pencil. Attached to the end of this document is some images of my book I took with my phone, which will hopefully make it a bit clearer.

Initial Reflection

Learning Needs

Already
Good

Improvement
Needs

Brainstorming

✓

Sketching 2D

✓

3D CAD

✓

Prototypes

✓

Analyses

✓

Scheduling

✓

Prepare presentations

✓

Writing Reports

✓

Would like to be better at:

1) Reading diagrams and interpreting them into the physical world

2) Using 3D software to model things effectively

Lily Rose

Age: 6

Personality: Hyper, excitable, loves bright colours and loud noises, very into dinosaurs

Goals: Lily is mostly going to the museum because her parents made her. She really, really wants to see a dinosaur though, and is not very interested in everything else.

Needs: She is very hyper, and loses attention quickly, so any exhibits should be flashy and bold to capture her attention. Since she is only six, she needs exhibits that are not too advanced for her to understand.

Frustrations: With a small attention span, Lily hates waiting for things. Since she is very small, she hates walking around long distances.

Pranz Cortez

Age: 19

Personality: Pranz is very nerdy, loving to delve into obscure topics online. He is also very talkative, loving to tell anybody around him the facts he knows.

Goals: To add new information to his repertoire. Would also love to critique the museum on the accuracy of some exhibits.

Needs: Pranz knows a lot, so he needs more obscure ^{or novel} information to keep him engaged.

Frustrations: Simplistic exhibits using basic language to cater to the average joe.

ENGEN 112

18 JUL 8:1

132-132

Mr Stan

Age: 73

Personality: Mr Stan is a fairly reserved fellow. He is getting older and is slow to do most anything.

Goals: He just came off a cruise and is sightseeing the city, so he is looking for things that will impress those back home.

Needs: Places to rest regularly, as he cannot stay on his feet all day.

Frustrations: When facilities to recuperate are far in between, such as places providing food, water, toilets.

ENGGEN 115

28 JUL 2026

By

LPI ast

Tutor

1/1

User Requirements

- Engaging
 - Exhibit should be interesting and excite the viewers
- Informative
 - Exhibit should effectively convey information to a broad audience
- Safe
 - Public should be kept well clear of moving parts & dangerous situations.

Constraints / Controls

- Museum rules on ramp clearance
 - 3m above level of door platform ^{over ramp} to be kept clear when not in operation
 - Crate must remain directly above ramp

System Requirements

- Crate dimensions
 - 3m x 3m x 3m max
 - Weight up to 2,000 kg
- Slope dimensions
 - 3.4m rise
 - 10m length
 - 10m width
- Platforms
 - Both flat
 - 7m long
 - 10m wide
 - Crate can be left at most 2m upwards.

Concepts

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05 AUG 2026

By IS (0.5/1)
Tutor



↑ Visitor lifts crate with pulleys

Top view



← Large lever so visitors can pull up crate (counterweight on other side)

Side



Gears in use

Potential vs kinetic energy



Cables elevate crate

User Requirements

Engaging

- The exhibit should capture the attention of the visitor
- It should be novel in some way, to separate it from other museums
- It should be of interest to a broad range of visitors

Informative

- Teaches the visitor about one or more engineering concepts
- Information should be simple enough for a layman to understand

Complete

- The museum is seeking a complete solution
 - ↳ Should include displays, components
 - ↳ Should take account of crowd control measures

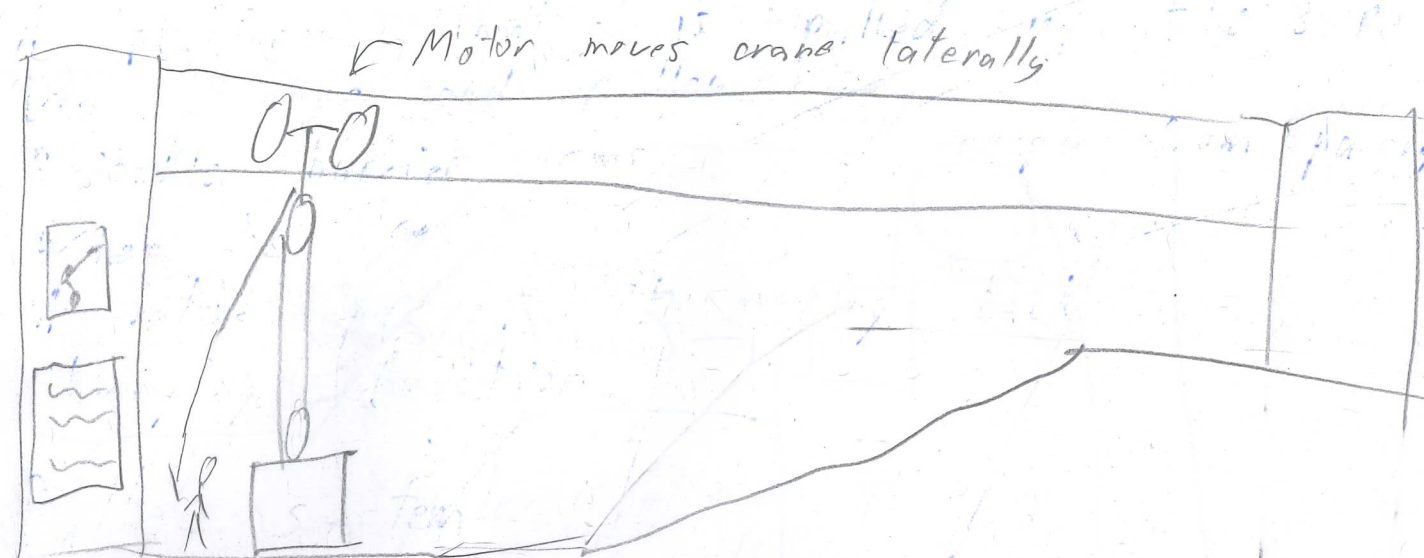
Constraints/Controls

- Museum rules on ramp clearance
 - ↳ Over ramp, 3m below level of upper platform to be kept clear when dormant
 - ↳ Crate must remain directly over ramp in operation
- Safety regulations
 - ↳ Modern safety regulations may require some concessions to be made.
- Dimensions
 - ↳ Dimensions of box and ramp are fixed in the brief and cannot be changed.

System Requirements

- Public - Extra care must be taken with the public around.
 - ↳ Not all are able/willing to read signage
- Environment - Ramp appears to be outdoors, placing machinery at higher risk of corrosion and deterioration.
- Cost - While not explicitly stated, museums typically have tight finances, relying on grants and other income sources.
 - ↳ Project should use cheaper materials and methods of construction where possible.

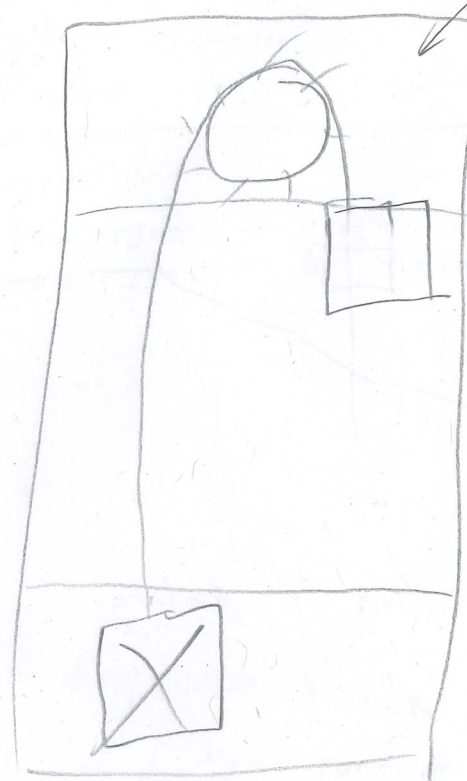
Crane System



- Lifting mechanism includes a ratchet to prevent crate being accidentally dropped 
- Large structure provides spectacle and draws attention.

Pulley

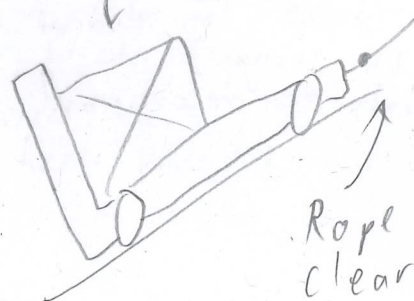
Top-down view



Wheel with spokes visitors
can push around
↳ Teaches concept of torque

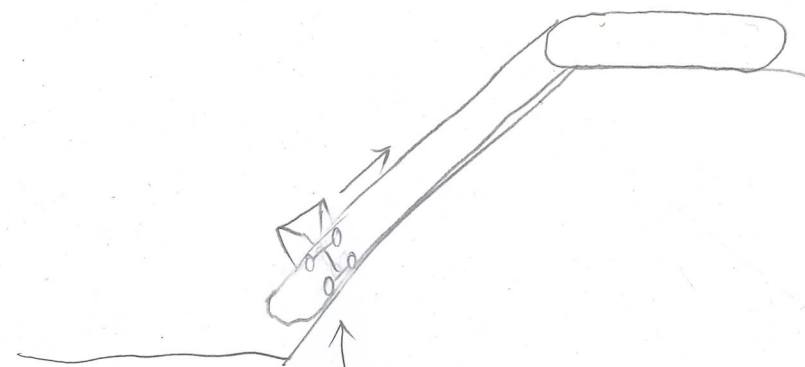
Counterweight

Crate
↳ Would need some sort
of trolleys to pull up ramp



Rope can be disconnected to
clear ramp

Side rails



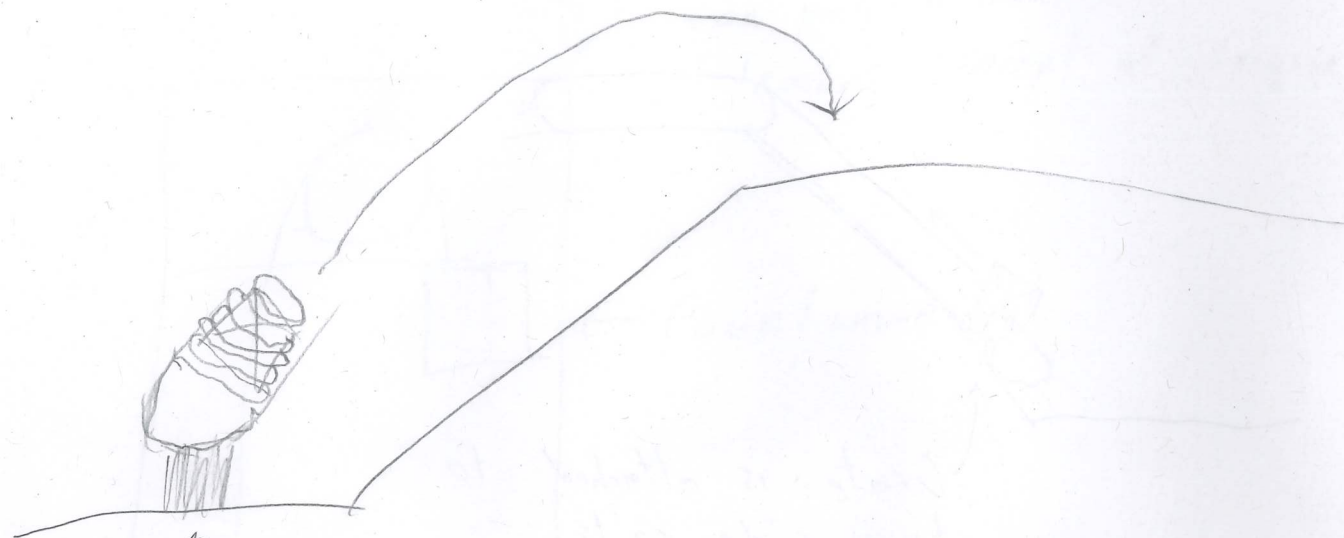
Crate is attached to
two side rails
↳ These rails energise an electric cart which
will pull the crate to the top.



Motor in hinge to lift rails

After crate has been moved, rails
change to vertical to clear the ramp.

Launch



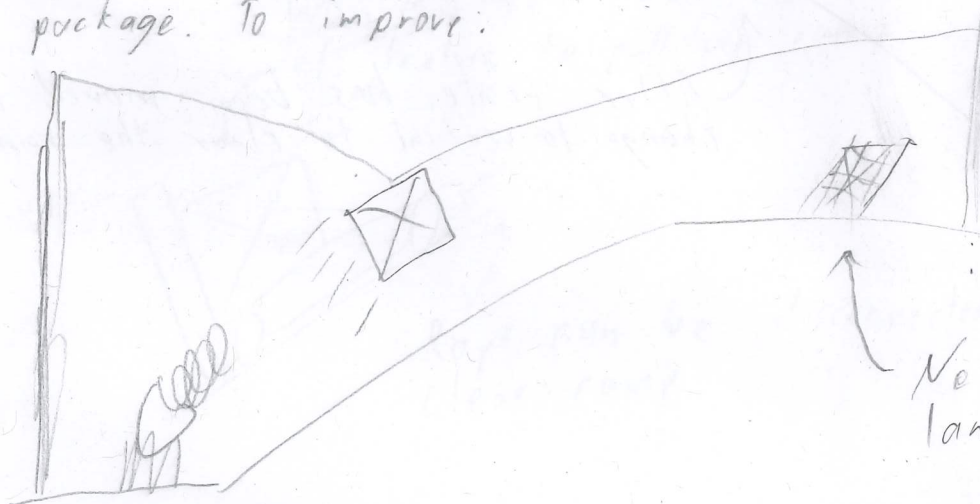
Electromagnetic railgun

↳ Teaches forces of magnetic repulsion

↳ Shows how electric and magnetic fields are intertwined

↳ May be difficult to convey an unintuitive topic

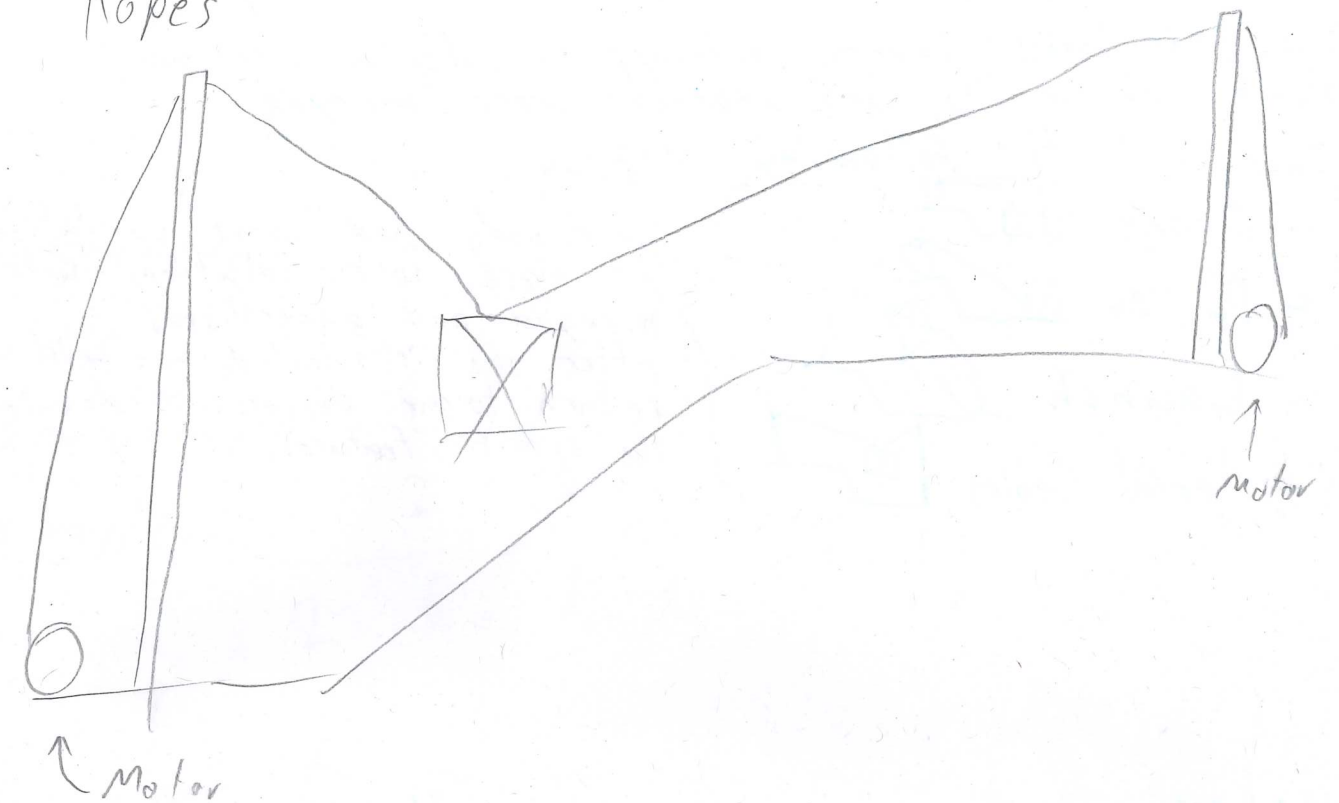
This has high risks of injuring others and the package. To improve:



Net will soften landing

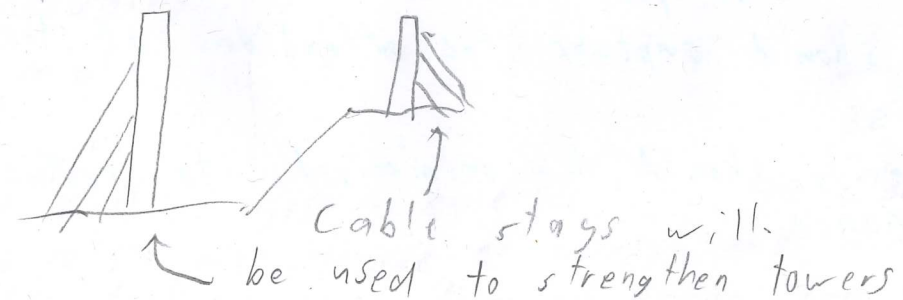
Ropes will guide trajectory

Ropes



Ropes elevate and guide crate to the top of the ramp

Towers must carry a significant load



Concept Selection

- The team had several overlapping design concepts, those similar to one another were merged.
- This left the following designs

- Crane 
- Pulley 
- Launch 
- Aerial Cables 

Everybody had some variation of these, so the solutions were merged and generalised. After one is selected, we will return to our designs to debate the specific features.

A MART analysis was completed on these using the following categories:

- Informative - The exhibit should serve to teach guests about an engineering topic
- Safe - The exhibit should have a high degree of safety due to the public environment it is in
- Engaging - It should capture the attention of the public
- Simple - Complexity should be minimised to reduce maintenance

	More Important			
	Informative	Safe	Engaging	Simple
Informative	-	2	1	1
Safe	0	-	0	0
Engaging	1	2	-	2
Simple	1	2	0	-
	2	6	1	3

0-4	Crane	Pulley	Launch	Cables		Crane	Pulley	Launch	Cables
Informative	3	4	3	2		6	8	6	4
Safe	3	4	0	1	→	18	24	0	6
Engaging	3	2	4	3	Weighted	3	2	4	4
Simple	1	4	2	3		3	12	6	9
						30	46	16	23

The pulley system emerges as the clear winner here, largely due to its high degree of safety and simplicity. (sourced from industry concepts)
Extra features were debated among team members to add these include:

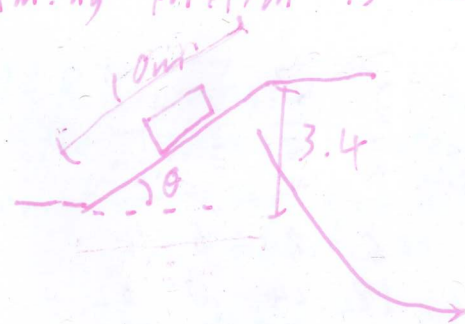
- Barriers to prevent pedestrians walking in front of box
- A system of pulleys and levers to reduce the load such that a museum visitor could pull it up the hill, increasing engagement.
- Wheels of varying friction will be used to teach the public about forces of resistance.
- A braking system will be added to improve safety in the event of a pulley failure.

Pulley System

- Box weighs 2,000 kg max
- Aim: 100x reduction to 20 kg ^{worth of force} so visitors can pull box up ramp.
- Considerations:
 - ↳ Should not fall back if let go
 - ↳ Rope must be removable to keep ramp clear

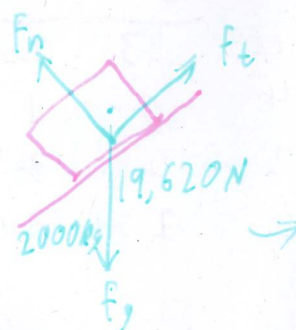
Force Calculations:

↳ Assuming friction is negligible



$$\theta = \sin^{-1}\left(\frac{3.4}{10}\right)$$

$$\theta = 19.88^\circ$$



$$\cos \theta_2 = \frac{F_t}{F_g}$$

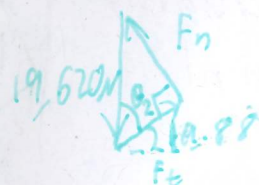
$$19620 \cos(70.12^\circ) = F_t$$

$$F_t = 6,672 \text{ N}$$

Force required

20kg worth desired = 196.2 N

Minimum 34x reduction



$$\theta_2 = 90 - 19.88^\circ$$

$$= 70.12^\circ$$

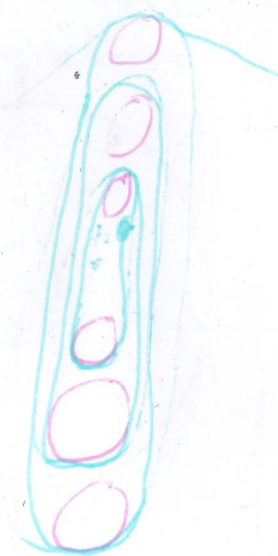
Force provided via capstan and pulley

Capstan sits on 2x10m platform

- Rope wraps around 0.5m radius
- Participants push 3m radius spoke
- 6x force multiplication



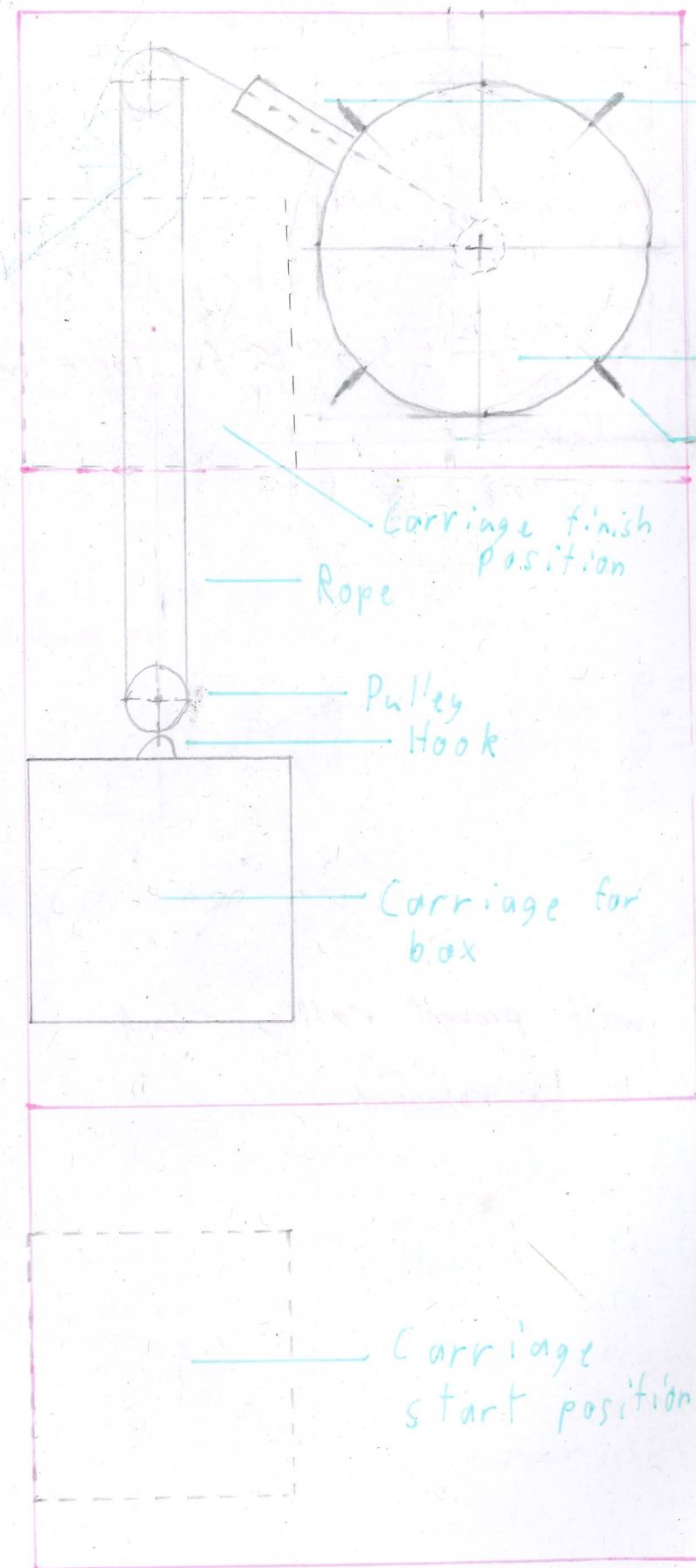
Pulley must then provide $\frac{3.4}{6} = 5.67 \approx 6x$ force multiplication
Can be done with 6 pulleys:



Ratchet on capstan will prevent rolling back
Hook will allow rope to be removed

Top-down view in operation

To Scale 1:100



Rope covering to prevent trips

Capstan

Handle (removable due to space limitations)

Carriage finish position

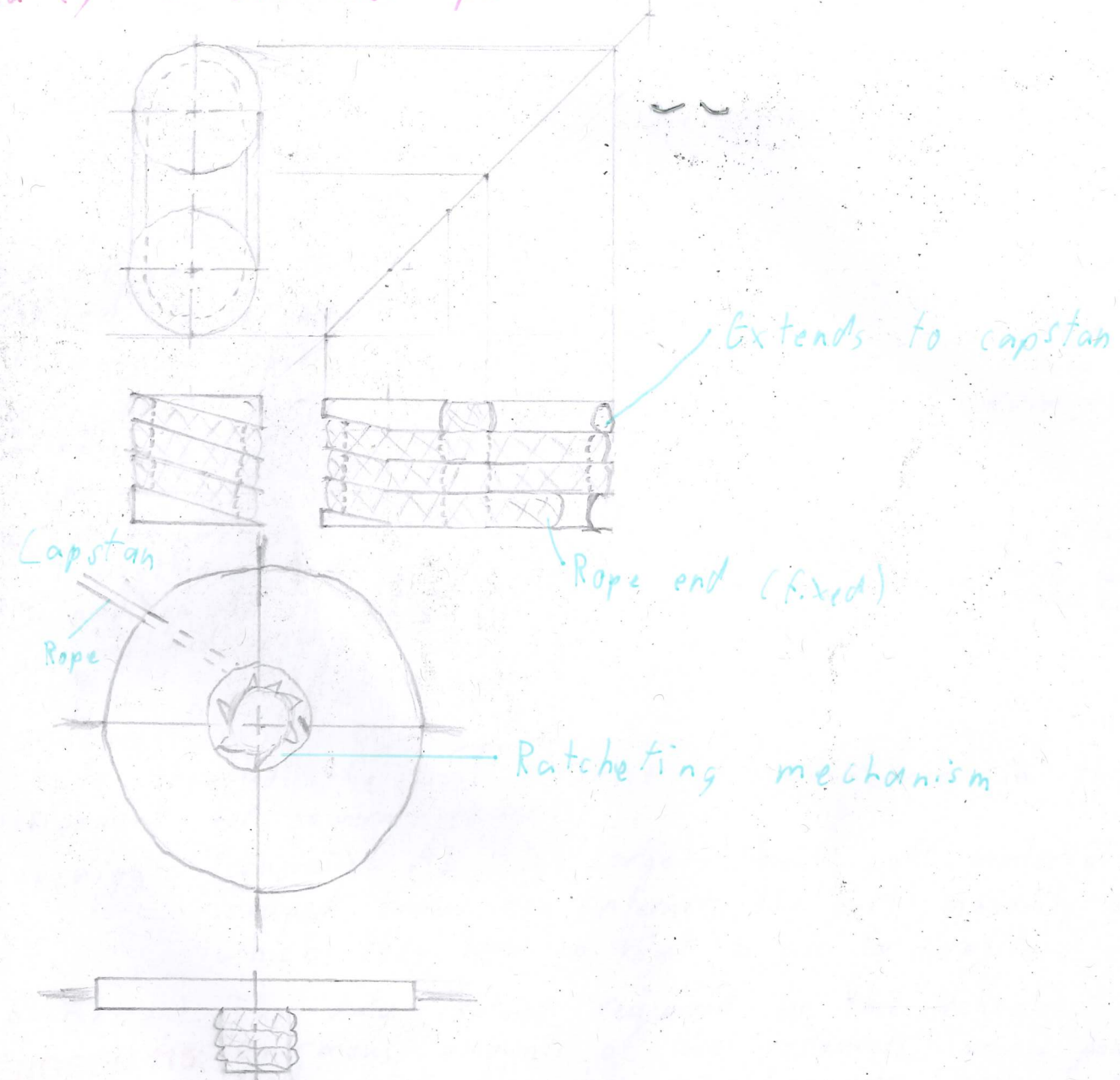
Rope

Pulley Hook

Carriage for box

Carriage start position

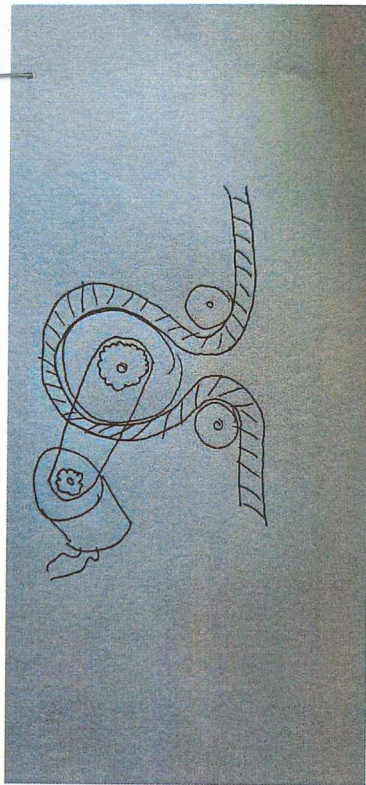
Pulley in shortened position



Extends to capstan

Rope end (fixed)

Ratcheting mechanism



AI image reflection

AI image generation can be incredibly effective during the early design process to gauge how an idea may look in real life. It is easy to use, quick, and cheap compared to the time and resources it would require to have somebody model it.

Its biggest drawback, and the reason I think AI is best suited to the early design stages, is it is inaccurate. AI frequently spits out nonsensical images that can have little to no bearing in reality. For example the image provided:

- The rope is not attached to anything
 - The text is nonsense
 - The frame holding the two smaller wheels is not attached to anything, and would realistically scrape against the large wheel
- In addition to this, there are several ethical/environmental considerations as outlined below.

Conscientious Objection

I have rejected the use of artificial intelligence in this assignment for several reasons:

Copyright: Generative AI is largely built off material scraped from the internet, the vast majority of which they have no right to use for profit.

Environment: The data centres required for this technology consume enormous amounts of local resources, largely power and water. All while providing very little back.

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12 AUG 2026

By

15 (1/1)

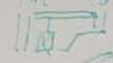
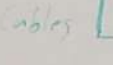
Tutor

e.g. The data centre planned for southland will employ 150 people, while its impact on our electricity grid is second only to Tiwai Point (which employs 1800 people, 1200 contractors, and supports an additional 13,000 jobs in the area).

Concerning Use Cases: The biggest detractor for me is what this technology is being used for. People are relying on it for accurate information when it consistently gets things wrong (I attempted to use it once for my work and it messed up the right hand rule). It is also being used to replace human creativity in *favour* of profits. On a more insidious level, AI has convinced people to take their own lives, and has been used to generate explicit images of real people, as well as CSAM. Until there is tighter regulation on these companies I doubt I will be using AI at all.

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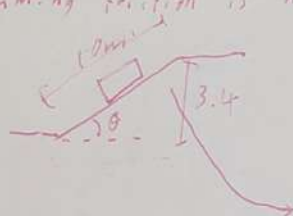
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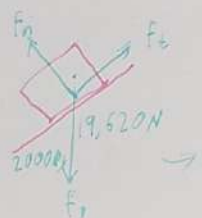
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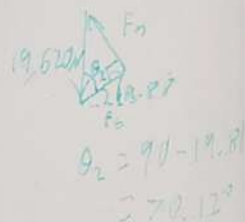
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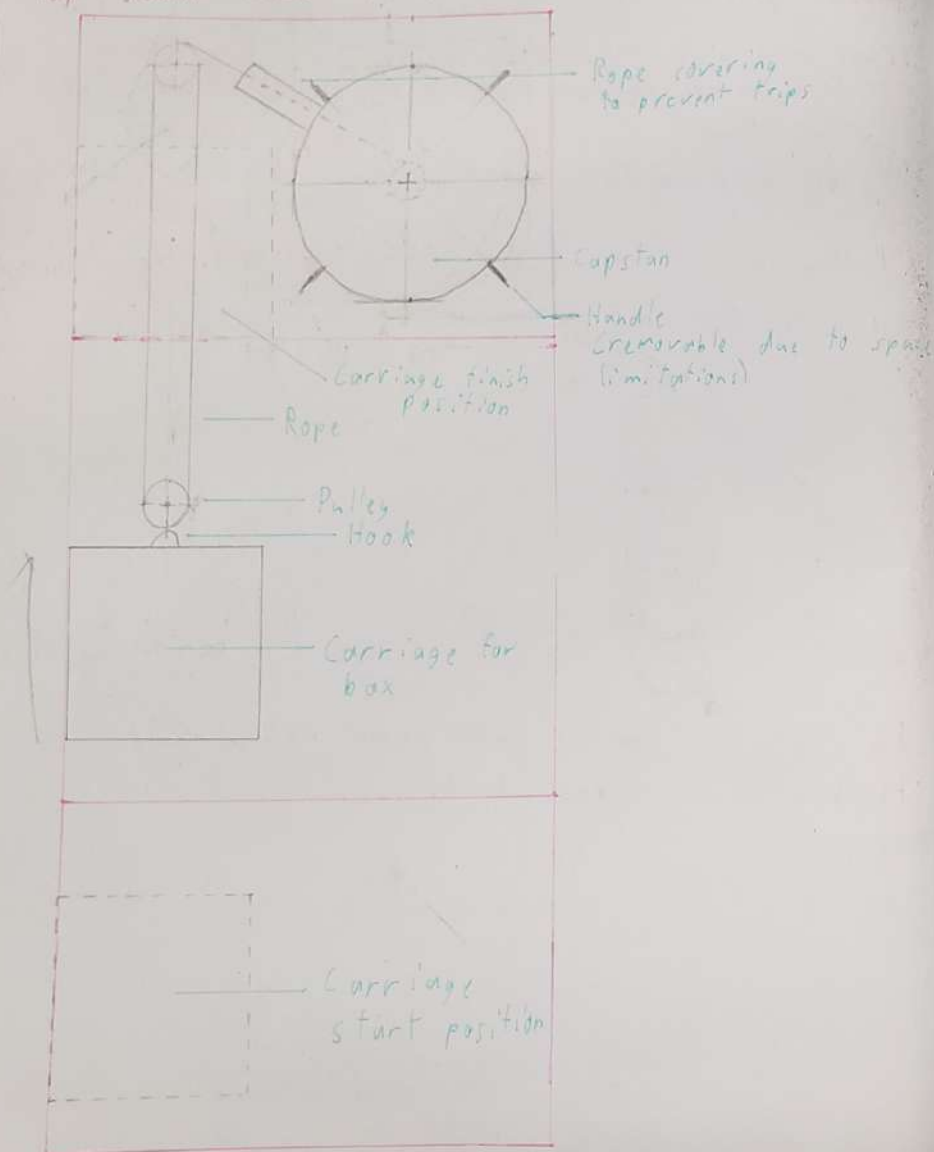
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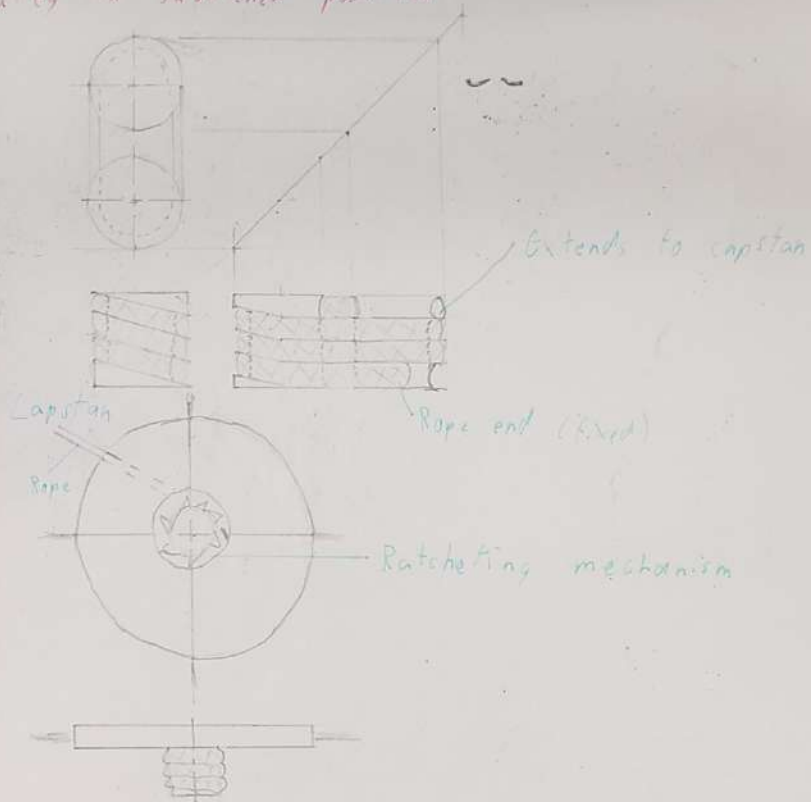
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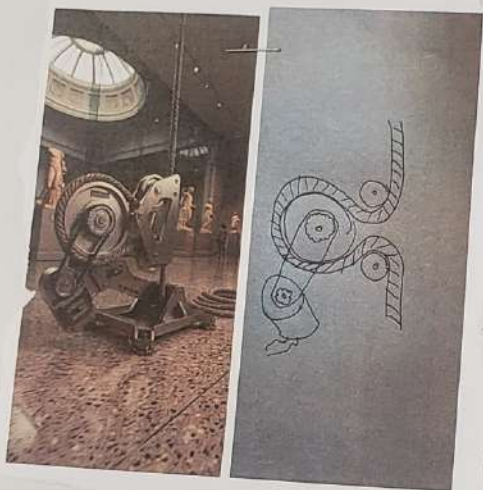
Top-down view in operation

To scale 1:100



Pulley in shortened position





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By 1.5 (1/1)

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